

Making Artistic PCBs with KiCad

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In this class we will be going over designing PCBs with KiCad.

We're not going to go over system design, component selection, or schematic design.

This is the typing class to a creative writing class

Class 1

- Learn what a PCB is and how it's constructed
- Run KiCad and learn about the tools:
 - Schematic Editor
 - Layout Editor
 - Learn about Gerber files
 - Gerber View
- Layout a simple board
 - Start with a schematic
 - Place components **

1. Components

A physical device that does something in a circuit

- LED
- Resistor
- Connector

2. LED

available in intermediate lighting. LEDs are high efficiency devices. LEDs have many applications, such as, strip lights, traffic signals, and digital devices.^[8]

LEDs have many advantages, such as lower power consumption, smaller sizes, and longer life. However, the disadvantages of LEDs are that they are not AC power, the electrical supply must be at a constant temperature.

LEDs were first developed by Henry Joseph Plesch and his team at General Electric. Round LEDs emit yellow, light

First produced

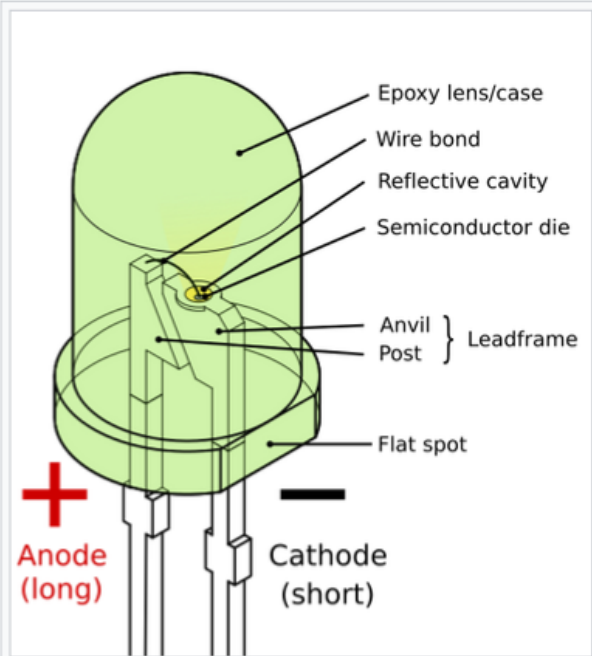
Oleg Losev (1927)^[2]
James R. Biard (1961)^[3]
Nick Holonyak (1962)^[4]

Pin names

Anode and cathode

Electronic symbol





Parts of a conventional LED. The flat bottom surfaces of the anvil and post embedded inside the epoxy act as anchors, to prevent the

Birthday mode (l

- ☐ Disabled
- ☒ Enabled

[Learn more about mode](#)

Text

- ☐ Small
- ☒ Standard
- ☐ Large

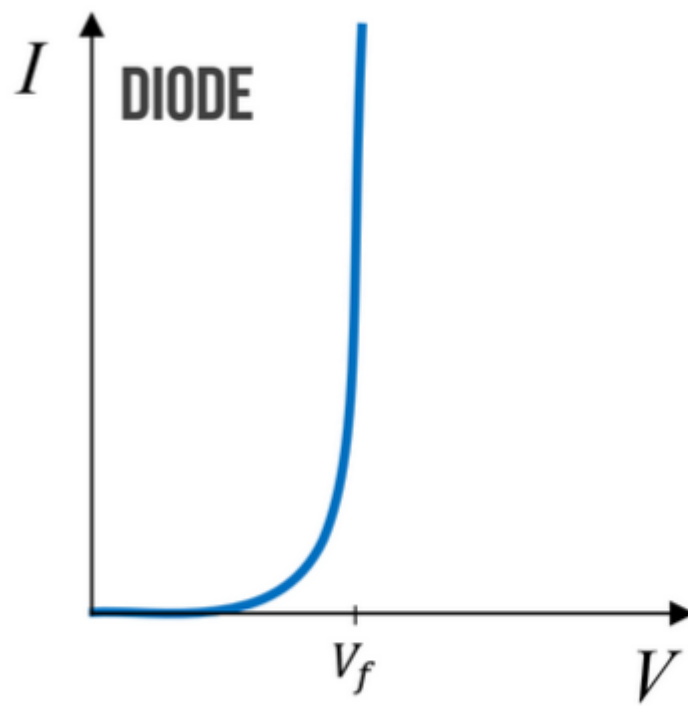
Width

- ☒ Standard
- ☐ Wide

Color (beta)

- ☐ Automatic
- ☒ Light
- ☐ Dark

An LED is a light emitting diode. It converts current to light



that after the applied voltage V exceeds the forward voltage V_f of the diode, (closed switch). Image made in PowerPoint.

3. Resistor

Resistor $V = IR$

4. Connector

5. What is a circuit?

A bunch of physical components that are connected so that they do something.

6. What is a PCB?

Sandwich with a fiberglass core. Layer stackup * Silk Screen * Solder mask * Copper foil * epoxy * Fiber glass core * epoxy * Copper foil * Solder mask * Silk screen

7. Through hole vs Surface Mount

Kicad

8. Schematic Editor